

8.012 Problem Set 4

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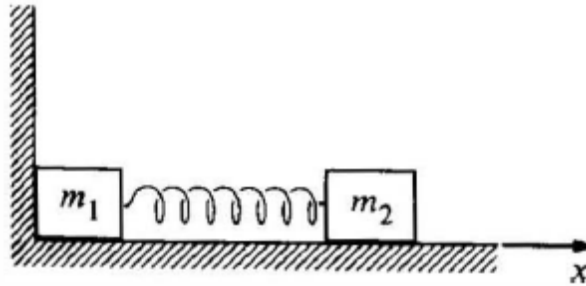
December 2025

Problem 1

Kleppner & Kolenkow, Problem 3.7

3.7 A system is composed of two blocks of mass m_1 and m_2 connected by a massless spring with spring constant k . The blocks slide on a frictionless plane. The unstretched length of the spring is l . Initially m_2 is held so that the spring is compressed to $l/2$ and m_1 is forced against a stop, as shown. m_2 is released at $t = 0$.

Find the motion of the center of mass of the system as a function of time.



As m_2 is released, the Spring Force starts to show its action.

Let the Wall be regarded as the position $x = 0$. The Right side is considered as Positive x .

Force on m_1 :- The Spring Forces pushes it to the Left side. The wall provides an Equal Normal Force which pushes it to the Right side. Hence the net Force is Zero. This leads to Acceleration of m_1 till the time the Spring is remained Compressed to be

$$\ddot{x}_{m_1} = 0$$

But after the Initial Compression state is over (say at time t_c), the Acceleration is given by

$$\ddot{x}_{m_1} = \frac{1}{m_1}k(x_{m_2} - x_{m_1} - l) \text{ if and only if } x_{m_1} \neq 0 \text{ after time } t_c$$

Force on m_2 :- The Only Force acting is the Spring Force.
Acceleration on m_2 is given by

$$\ddot{x}_{m_2} = \frac{1}{m_2}k(l - (x_{m_2} - x_{m_1}))$$

So at time $t < t_c$, the equation of motions are

$$\begin{aligned} \ddot{x}_{m_1} &= 0; \dot{x}_{m_1} = 0; x_{m_1} = 0 \\ \ddot{x}_{m_2} + \frac{k}{m_2}x_{m_2} &= \frac{kl}{m_2} \\ \implies x_{m_2} &= c_1 \cos\left(\sqrt{\frac{k}{m_2}}t\right) + c_2 \sin\left(\sqrt{\frac{k}{m_2}}t\right) + l \end{aligned}$$

Solving for c_1 and c_2 ,

$$\begin{aligned} x_{m_2 < initial} &= l/2 \implies c_1 = -\frac{l}{2} \\ \dot{x}_{m_2 < initial} &= 0 \implies c_2 = 0 \end{aligned}$$

This Gives us

$$x_{m_2} = -\frac{l}{2} \cos\left(\sqrt{\frac{k}{m_2}}t\right) + l \text{ when } t < t_c$$

Position of Center of Mass in time $t < t_c$ is given by

$$\begin{aligned} &\frac{(m_1)(0) + (m_2)(-\frac{l}{2} \cos\left(\sqrt{\frac{k}{m_2}}t\right) + l)}{m_1 + m_2} \\ \implies x_{C.M.} &= \frac{m_2 l}{m_1 + m_2} \left(1 - 0.5 \cos\left(\sqrt{\frac{k}{m_2}}t\right)\right) \\ \implies \dot{x}_{C.M.} &= \frac{m_2 l}{m_1 + m_2} \left(0.5 \sqrt{\frac{k}{m_2}} \sin\left(\sqrt{\frac{k}{m_2}}t\right)\right) \end{aligned}$$

Note that

$$t < t_c \implies x_{m_2} < l$$

So,

$$x_{m_2}(t_c) = l \implies t_c = \frac{\pi}{2} \sqrt{\frac{m_2}{k}}$$

After $t = t_c$, the block m_1 loses contact with the wall and the external normal force vanishes. Hence the net external horizontal force on the system is zero. Since internal forces do not affect the motion of the center of mass, the center of mass moves with constant velocity for $t > t_c$.

Now

$$\begin{aligned}\dot{x}_{C.M.}(t_c) &= \frac{m_2 l}{m_1 + m_2} \left(0.5 \sqrt{\frac{k}{m_2}} \right) \\ \dot{x}_{C.M.} &= \dot{x}_{C.M.}(t_c) = \frac{m_2 l}{m_1 + m_2} \left(0.5 \sqrt{\frac{k}{m_2}} \right) \text{ for } t > t_c \\ \Rightarrow x_{C.M.} &= x_{C.M.}(t_c) + \frac{m_2 l}{m_1 + m_2} \left(0.5 \sqrt{\frac{k}{m_2}} \right) (t - t_c) \\ &= \frac{m_2 l}{m_1 + m_2} + \frac{m_2 l}{m_1 + m_2} \left(0.5 \sqrt{\frac{k}{m_2}} \right) (t - t_c)\end{aligned}$$

Hence the **Position of the C.M. can be written as**

$$\begin{aligned}x_{C.M.} &= \frac{m_2 l}{m_1 + m_2} \left(1 - 0.5 \cos \left(\sqrt{\frac{k}{m_2}} t \right) \right) \text{ if } t \leq \frac{\pi}{2} \sqrt{\frac{m_2}{k}} \\ x_{C.M.} &= \frac{m_2 l}{m_1 + m_2} \left(1 + 0.5 \sqrt{\frac{k}{m_2}} \left(t - \frac{\pi}{2} \sqrt{\frac{m_2}{k}} \right) \right) \text{ if } t \geq \frac{\pi}{2} \sqrt{\frac{m_2}{k}}\end{aligned}$$

The Velocity and Acceleration can be calculated accordingly The center of mass accelerates for $t < t_c$ due to the external normal force exerted by the wall, and moves with constant velocity for $t > t_c$ once the system becomes isolated.

Problem 4

Kleppner & Kolenkow, Problem 3.14

3.14 N men, each with mass m , stand on a railway flatcar of mass M . They jump off one end of the flatcar with velocity u relative to the car. The car rolls in the opposite direction without friction.

a. What is the final velocity of the flatcar if all the men jump at the same time?

b. What is the final velocity of the flatcar if they jump off one at a time? (The answer can be left in the form of a sum of terms.)

c. Does case a or case b yield the largest final velocity of the flat car? Can you give a simple physical explanation for your answer?

a.

If v is the Final Velocity of the flatcar, by the Conservation of Linear Momentum,

$$(M)(v) + (Nm)(u + v) = (M + Nm)(0) \implies v = -\left(\frac{Nm}{M + Nm}\right)u$$

Hence

Final Velocity of the flatcar if all men jump at the same time is given by

$$\left(\frac{Nm}{M + Nm}\right)u \text{ Opposite to the Direction in which the men jumped}$$

b.

If v_r is the Velocity of the flatcar after r man men jumped, by the Conservation of Linear Momentum,

$$(M + (N - r)m)v_r + m(u + v_r) = (M + (N - (r - 1))m)v_{r-1}$$

$$\implies v_r - v_{r-1} = \frac{-mu}{M + (N + 1 - r)m}$$

Therefore,

$$\text{Final Velocity} = -mu \sum_{r=1}^N \left[\frac{1}{M + (N + 1 - r)m} \right]$$

Hence

Final Velocity of the flatcar if all men jump one by one is given by

$$mu \sum_{r=1}^N \left[\frac{1}{M + (N + 1 - r)m} \right] \text{ Opposite to the Direction in which the men jumped}$$

c.

Note that r varies from 1 to n in case b.

But the term of the Summation with $r = 1$ is exactly equal to the answer of case a.

All the terms in the Summation of case b.

**Since later jumps occur when the flatcar has smaller mass,
each jump produces a larger recoil.
Hence case (b) gives a larger final velocity.**

Problem 7

Kleppner & Kolenkow, Problem 3.18

3.18 A raindrop of initial mass M_0 starts falling from rest under the influence of gravity. Assume that the drop gains mass from the cloud at a rate proportional to the product of its instantaneous mass and its instantaneous velocity:

$$\frac{dM}{dt} = kMV,$$

where k is a constant.

Show that the speed of the drop eventually becomes effectively constant, and give an expression for the terminal speed. Neglect air resistance.

$$\frac{d}{dt}(VM) = F_{external} + \sum (u_{rel}\dot{m}) \text{ velocity of incoming/outgoing mass relative to the body}$$

$$\implies \dot{V}M + V\dot{M} = Mg + (0)(0) \implies \dot{V} = g - kV^2$$

$$\implies V = \sqrt{\frac{g}{k}} \tanh(t\sqrt{kg})$$

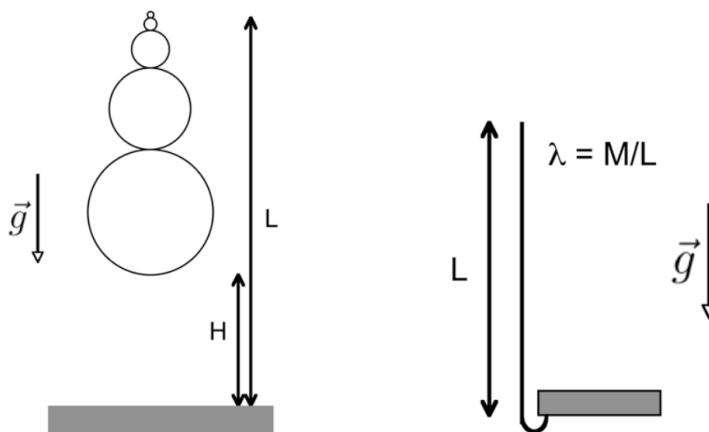
So

$$t \rightarrow \infty \implies V \rightarrow \sqrt{\frac{g}{k}}$$

Hence Terminal Speed is given by

$$V_{Terminal} = \sqrt{\frac{g}{k}}$$

Problem 9 : The Super-Astro-Blaster



9. The super-astro-blaster. [15 points]

In class we demonstrated how the astroblaster (a set of elastic bouncing balls released from rest and rebounding from the floor) can launch the smallest and topmost ball to considerable distance. Let's push this to the test. Assume we have n balls of mass (left figure above) $m_1, m_2, m_3, \dots, m_n$ where $m_1 \gg m_2 \gg m_3 \gg \dots > m_n$. The most massive ball is at bottom and starts at rest at a height $H = 1$ meter above the ground. Assume that all balls bounce elastically, that constant gravitational acceleration $\vec{g}$ points downward, and that air resistance and realistic limits in the elasticity of the balls can be ignored.

- (10 points) In terms of n , to what height does the top ball bounce? How many balls do you need to reach 1 kilometer?
- (5 points) How many balls do you need to achieve escape velocity? You will need to determine an expression for escape velocity from the Earth, which is possible (to within an order of magnitude) through dimensional analysis.

Just before the bottom ball strikes the ground, all balls have the same downward speed

$$v_0 = \sqrt{2gH}.$$

After bouncing elastically from the ground, the bottom ball reverses its velocity to $+v_0$. It then collides elastically with the next ball, which is still moving downward with speed v_0 .

In the limit $m_1 \gg m_2$, the lighter ball acquires approximately twice the relative velocity. Thus the velocity of the second ball after collision is

$$v_2 \approx 2v_0.$$

This process repeats successively for each ball due to the strong mass hierarchy.

Velocity Amplification

After the $(n - 1)$ successive elastic collisions, the top ball acquires speed

$$v_n \approx 2^{n-1} v_0.$$

a.

The maximum height reached by the top ball is

$$h_n = \frac{v_n^2}{2g} = \frac{(2^{n-1} v_0)^2}{2g} = 2^{2(n-1)} H$$

$$h_n = 4^{n-1}$$

With $H = 1$ m, we require

$$4^{n-1} \geq 10^3.$$

Taking logarithms,

$$n - 1 \geq \frac{\ln 10^3}{\ln 4} \approx 5,$$

so

$$n \approx 6.$$

Hence

6 balls are sufficient to reach 1 km

b.

The escape velocity from Earth is

$$v_{\text{esc}} \sim \sqrt{gR},$$

where R is the radius of the Earth.

Requiring

$$2^{n-1} v_0 \sim \sqrt{gR},$$

with $v_0 = \sqrt{2gH}$, we obtain

$$2^{n-1} \sim \sqrt{\frac{R}{2H}}.$$

For $R \sim 6 \times 10^6$ m and $H = 1$ m,

$$n \approx 12.$$

About 12 balls are required to reach escape velocity

Problem 10

10. A falling chain. [10 points]

You hold a chain of mass M , length L and uniform mass density λ from one end above a fixed table (right figure above). The chain falls under constant gravitational acceleration. Ignore any swinging or bouncing in the chain and assume that it moves purely in a vertical direction until it comes to rest hanging straight down below the table.

(10 points) What is the support force from the table, F_S , as a function of time after the chain is released?

Let $x(t)$ be the length of the chain that has come to rest on the table at time t . Since the chain falls freely under gravity,

$$x(t) = \frac{1}{2}gt^2 \quad \text{for } 0 \leq t \leq \sqrt{\frac{2L}{g}}.$$

The mass of the chain resting on the table at time t is

$$m(t) = \lambda x(t).$$

The support force exerted by the table has two contributions:

1. The weight of the chain already at rest on the table:

$$F_{\text{weight}} = m(t)g = \lambda g x(t).$$

2. The force required to stop the incoming chain elements. Each infinitesimal element arrives with velocity

$$v(t) = gt,$$

and the rate at which mass arrives at the table is

$$\dot{m} = \lambda v(t).$$

Hence the momentum flux is

$$F_{\text{momentum}} = \dot{m}v = \lambda v^2(t).$$

Therefore, the total support force is

$$F_S(t) = \lambda g x(t) + \lambda v^2(t).$$

Substituting $x(t) = \frac{1}{2}gt^2$ and $v(t) = gt$,

$$F_S(t) = \lambda g \left(\frac{1}{2}gt^2 \right) + \lambda (gt)^2 = \frac{3}{2}\lambda g^2 t^2.$$

$$F_S(t) = \frac{3}{2}\lambda g^2 t^2 \quad \left(0 \leq t \leq \sqrt{\frac{2L}{g}}\right)$$

After the entire chain has landed on the table, the support force becomes constant:

$$F_S = Mg.$$